Experiment Report

# Experiment Report Name

Author

Month DD, YYYY

# Document Control

| | |
|---|---|
| **Version** | [Version] |
| **Prepared By** | [Prepared by / team] |
| **Reviewers** | [Reviewer roles or names] |
| **Date Prepared** | [Date prepared] |
| **Reporting Window** | [Reporting start date] to [Reporting end date] |
| **Status** | [Draft / In review / Final] |

# Purpose

[Describe what this document records, including the experiment objective, execution, analysis, and decision it supports. Note the product surface and intended scope.]

[Explain why this record is being maintained, who should use it, and how it supports auditability, replication, or future decision-making.]

# Business Context

[Describe the company, product, user journey, and observed problem or opportunity. Summarize the evidence that motivated the experiment and where the issue appears in the experience.]

[Explain why this test surface and intervention were selected, including expected value, operational risk, reversibility, and any important alternatives that remained unchanged.]

# Experiment Summary

| Field | Details |
|---|---|
| **Experiment Name** | [Experiment name] |
| **Experiment Key** | [Experiment key] |
| **Owner Team** | [Owner team] |
| **Business Owner** | [Business owner] |

| | |
|---|---|
| **Product Surface** | [Product surface] |
| **Primary Objective** | [Primary objective] |
| **Control (Variant A)** | [Describe the control experience] |
| **Treatment (Variant B)** | [Describe the treatment experience] |
| **Allocation** | [Variant allocation] |
| **Unit of Randomization** | [Randomization unit] |
| **Audience** | [Eligible audience] |
| **Exclusions** | [Traffic or user exclusions] |
| **Start Date** | [Start date] |
| **End Date** | [End date] |
| **Planned Runtime** | [Planned runtime] |
| **Actual Runtime** | [Actual runtime] |

## Design Notes

[Describe the experimental design and the single change being tested. Explain how the design supports causal interpretation and which elements remained unchanged.]

[Describe eligibility, assignment, exclusions, runtime, and other choices made to reduce contamination or bias. Note any platform, geography, or device constraints.]

## Hypothesis

[Write an if/then hypothesis naming the control, treatment, audience, expected metric direction, and causal rationale.]

## Success Criteria

[Define the primary success metric, guardrail metrics, decision thresholds, and the conditions required to declare a winner or proceed.]

| Metric | Target/Rule |
|---|---|
| **Primary Metric** | [Define the primary metric threshold and statistical decision rule] |
| **Guardrail 1** | [Define guardrail metric and acceptable threshold] |
| **Guardrail 2** | [Define guardrail metric and acceptable threshold] |
| **Guardrail 3** | [Define guardrail metric and acceptable threshold] |
| **Decision Rule** | [State the full launch, hold, or rollback rule] |

# Pre-Launch Validation

- [Confirm that only the intended treatment differs between variants.]
- [Confirm assignment persistence, randomization unit, and allocation logic.]
- [Validate exposure, outcome, and guardrail instrumentation.]
- [Document traffic exclusions and data-quality filters.]
- [Record staging, A/A, or other pre-launch validation results.]

# Sample and Exposure Summary

[Summarize achieved sample size, exposure balance, runtime, and any sample-ratio or eligibility findings. State whether the data are suitable for analysis.]

| Metric | Variant A (Control) | Variant B (Treatment) |
|---|---|---|
| **[Sample metric 1]** | [Variant A value] | [Variant B value] |
| **[Sample metric 2]** | [Variant A value] | [Variant B value] |
| **[Outcome count]** | [Variant A value] | [Variant B value] |
| **[Exposure or allocation share]** | [Variant A value] | [Variant B value] |
| **Sample Ratio Check** | [Pass / Fail] | [Pass / Fail] |

# Primary Outcome

| Measure | Variant A | Variant B | Absolute Delta | Relative Delta |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| **[Primary outcome metric]** | [Variant A value] | [Variant B value] | [Absolute delta] | [Relative delta] |
| **95% Confidence Interval** | [N/A if not applicable] | [N/A if not applicable] | [Confidence interval] | [N/A if not applicable] |
| **Two-Sided p-value** | [N/A if not applicable] | [N/A if not applicable] | [p-value] | [N/A if not applicable] |
| **Outcome Narrative** | [N/A] | [N/A] | [Describe effect and practical significance] | [N/A] |

[Summarize the primary metric result, uncertainty, statistical significance, and practical effect size. Avoid claims beyond the experiment scope.]

# Guardrail Metrics

| Guardrail Metric | Variant A | Variant B | Delta | Status |
|---|---|---|---|---|
| **[Guardrail metric 1]** | [Variant A value] | [Variant B value] | [Delta] | [Pass / Fail] |
| **[Guardrail metric 2]** | [Variant A value] | [Variant B value] | [Delta] | [Pass / Fail] |
| **[Guardrail metric 3]** | [Variant A value] | [Variant B value] | [Delta] | [Pass / Fail] |
| **[Guardrail metric 4]** | [Variant A value] | [Variant B value] | [Delta] | [Pass / Fail] |

[Summarize each guardrail result, note any threshold breaches, and explain whether the risk profile supports the proposed decision.]

# Segment Results

[Describe which segments were pre-specified, why they matter, and whether the experiment was powered for subgroup comparisons. Distinguish confirmatory from directional analysis.]

| Segment | Control Value | Treatment Value | Absolute Delta | Interpretation |
|---|---|---|---|---|
| **[Segment 1]** | [Control value] | [Treatment value] | [Absolute delta] | [Interpret result] |
| **[Segment 2]** | [Control value] | [Treatment value] | [Absolute delta] | [Interpret result] |
| **[Segment 3]** | [Control value] | [Treatment value] | [Absolute delta] | [Interpret result] |
| **[Segment 4]** | [Control value] | [Treatment value] | [Absolute delta] | [Interpret result] |
| **[Segment 5]** | [Control value] | [Treatment value] | [Absolute delta] | [Interpret result] |

[Summarize the most decision-relevant segment patterns and plausible mechanisms. Note uncertainty, multiplicity, or sample-size caveats.]

## Data Quality and Limitations

- [List concurrent launches, operational events, or external factors that could affect interpretation.]
- [Describe traffic-mix or eligibility changes during the experiment.]
- [Document logging, pipeline, or warehouse incidents and their resolution.]
- [State platform, geography, audience, or time-window limitations.]
- [Note treatment-specific risks, policy constraints, or generalizability limits.]

[Explain how these limitations affect the recommendation and clearly define where the evidence should and should not be generalized.]

## Interpretation

[Connect the observed results to the hypothesis and proposed mechanism. Explain whether the effect is statistically credible, practically meaningful, and consistent across key measures.]

[Translate the result into business or product impact, including expected upside, cost, reversibility, and remaining uncertainty. State the recommended decision posture.]

## Decision Log

| Item | Decision |
|---|---|
| **Final Decision** | [State final decision and scope] |
| **Decision Date** | [Decision date] |
| **Approvers** | [Approver roles or names] |
| **Rollout Type** | [Describe rollout or implementation approach] |
| **Rollback Trigger** | [Define rollback threshold or stop condition] |

| | |
|---|---|
| **Follow Up** | [Describe required follow-up validation] |

# Post Test Actions

1. [Describe the rollout or implementation sequence.]
2. [Define post-launch monitoring metrics, cadence, and duration.]
3. [List replication or follow-up experiments.]
4. [Identify additional analysis or product opportunities.]
5. [Document archival, reporting, and ownership requirements.]

# Appendix A: Metric Definitions

- **[Metric name 1]**: [Define numerator, denominator, eligibility, event source, and calculation.]
- **[Metric name 2]**: [Define numerator, denominator, eligibility, event source, and calculation.]
- **[Metric name 3]**: [Define numerator, denominator, eligibility, event source, and calculation.]
- **[Metric name 4]**: [Define numerator, denominator, eligibility, event source, and calculation.]
- **[Metric name 5]**: [Define numerator, denominator, eligibility, event source, and calculation.]

# Appendix B: Analyst Notes

- [Document the statistical method, confidence level, and software or query version used.]
- [Record stopping rules, interim reads, or deviations from the analysis plan.]
- [Summarize data reconciliation, backfills, and reviewer sign-off.]
- [Add any remaining assumptions, caveats, or provenance notes.]